System Validation Report

Execution of the functional test cases TC_01–TC_24 from Online_Exam_Portal_Test_Case_Plan.pdf.

1. Summary

Test cases	Pass	Partial	Not executed	Fail
24	22	1	1	0

Validation was performed against the running backend and database using the automated API test suite (57 tests) plus HTTP smoke checks. The browser UI was not exercised, so TC_11 (on-screen timer) is marked Partial and TC_24 (responsive layout) is Not executed; these two need a manual browser check before sign-off. No defects were found.

2. Environment

Item	Value
Project	Online Exam Portal (github.com/satvikkalyanpur/Online-Exam-Portal), commit f677021
Date of execution	06 November 2026
Runtime	Node.js v22.23.4, tsx test runner (node:test), Prisma 7.10.1
Database	MariaDB 10.11.14 (the project documents MySQL 8; the app uses the MariaDB driver adapter, which supports both)
Test data	Seeded with src/seed.ts (4 exams, 12 questions, 2 finished attempts, 4 accounts)
Test command	npm test (tsx --test --test-concurrency=1 "test/*.test.ts")

Environment note: the Prisma schema-engine binary could not be downloaded in the execution sandbox, so the database tables were created from database/schema.prisma using equivalent hand-written DDL instead of 'prisma db push'. The Prisma client was generated normally and the seed script ran unchanged.

3. Test case results

ID	Module	Verified	Actual result	Method	Test result
TC_01	Project Setup	Application starts and pages load	Server started with npm start equivalent; GET / returned 200; client production build succeeded (vite, 0 errors); typecheck (tsc) clean.	HTTP smoke + build	Pass
TC_02	User Login Authentication	Valid login	US01 test: valid credentials return a token and the role.	Automated (US01)	Pass
TC_03	User Login Authentication	Invalid credentials rejected	US01 tests: wrong password and unknown email give 401 with the same message; manual curl returned 401 "Invalid email or password."	Automated (US01) + curl	Pass
TC_04	Role-Based Access Control	Role-restricted functions	US02 tests: student/teacher/admin each denied features of the other roles; no-token requests rejected.	Automated (US02)	Pass
TC_05	View Available Exams	Student sees available exams	US03 tests: published exams listed with subject, schedule, duration and question count; drafts hidden.	Automated (US03)	Pass
TC_06	Create Examination	Admin creates exam	US04 tests: exam created with all fields; invalid details rejected.	Automated (US04)	Pass
TC_07	Question Bank	Add question	US05 tests: question added (linked or unlinked); invalid questions rejected.	Automated (US05)	Pass
TC_08	Question Bank	Edit question	US06 tests: edit saved; marked question wording can be fixed but not answer/marks.	Automated (US06)	Pass
TC_09	Question Bank	Delete question	US07 tests: deletion removes question; a question that has been marked cannot be deleted.	Automated (US07)	Pass

ID	Module	Verified	Actual result	Method	Test result
TC_10	Attend Examination Online	Start and answer exam	US08/US09/US10 tests: attempt starts, questions returned without answers, answers accepted, other students blocked.	Automated (US08–US10)	Pass
TC_11	Examination Countdown Timer	Timer reflects configured duration	Backend deadline (expiresAt) verified against the exam duration and window (US09, US13). The on-screen ticking countdown was not observed because no browser was available.	Automated (API only)	Partial – UI not observed
TC_12	Automatically Save Exam Answers	Answers saved during attempt	US10 tests: answer saved on selection, updated on change, retained on resume.	Automated (US10)	Pass
TC_13	Navigate Between Questions	Move between questions	US11 test: answers to several questions are kept independently while moving between them (state level). Next/Previous button behaviour was not observed in a browser.	Automated (US11)	Pass (data level)
TC_14	Restrict Exam Access Before Scheduled Time	Future exam blocked	US12 tests: unopened exam listed but cannot be started; can start once open; closed window blocked.	Automated (US12)	Pass
TC_15	Automatically Submit on Expiry	Auto-submit on timeout	US13 tests: expired attempt submitted even when the browser is closed; late answers refused; attempt never runs past the window.	Automated (US13)	Pass
TC_16	Automatically Evaluate Objective Questions	Scoring	US14 test: submitting marks answers and stores the score against the key.	Automated (US14)	Pass
TC_17	View Examination Results	Student result	US15 tests: no result before submit; result shown for the right exam; results private; correct answers hidden until exam closes.	Automated (US15)	Pass
TC_18	Student Performance Reports	Report view	US16 test: admin sees the attempt in the performance report, filtered by exam; non-admins denied.	Automated (US16)	Pass
TC_19	Examination Statistics	Statistics	US17 test: statistics computed from stored results; non-admins denied.	Automated (US17)	Pass
TC_20	Secure Credentials and Exam Data	Protected data	US18 tests: security headers/no-cache, bcrypt password hashes, lockout after repeated failures, refused requests logged.	Automated (US18)	Pass
TC_21	Data Consistency During Network Loss	Resend after loss	US19 tests: resent answers/submissions never duplicate; answer racing with submit is never stored unmarked; small post-deadline delay tolerated. Simulated at API level (retries), not a real browser disconnect.	Automated (US19)	Pass (API level)
TC_22	Backup and Recovery of Exam Data	Backup/restore	US20 tests: backup holds exams/results; restore recovers deleted data exactly; damaged file refused with no change; admin only.	Automated (US20)	Pass
TC_23	Maintain Activity Logs	Logging	US21 tests: login, exam start, submit and exam creation logged; filter/search; admin only.	Automated (US21)	Pass
TC_24	Responsive and User-Friendly Interface	Desktop/narrow viewport	Not executed. Requires a browser at desktop and mobile widths; none was available in the test environment.	Manual – pending	Not executed

4. Defects

None raised. Observed behaviour matched expected results for every executed case.